Partha Pratim Bhakat

Generative AI Engineer | Agentic AI Engineer

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Technical Skills

Languages: Python, Java, SQL, HTML

AI & Machine Learning: Generative AI, Hugging Face, LLMs, Agentic AI, Prompt Engineering, LangChain, LangGraph

Data & Web Engineering: NumPy, Pandas, Matplotlib, Seaborn, MySQL

Tools & Platforms: Git, GitHub, VS Code

Projects

MathsGPT: AI Math Problem Solver

Python, LangChain, Streamlit, Groq | math-chatbot-ppb.streamlit.app

- Developed an interactive mathematical AI assistant utilizing a LangChain Zero-Shot ReAct Agent to seamlessly orchestrate Wikipedia search, computational, and reasoning tools.
- Engineered a specialized problem-solving pipeline integrating LLMMathChain model via Groq API, delivering rapid, step-by-step solutions to complex mathematical queries.
- Built a highly responsive conversational user interface using Streamlit, facilitating real-time inference and providing structured, point-wise logical deductions.

Nexus Autonomous Agents: Multi-Agent AI Assistant

Python, LangChain, LangGraph, Streamlit, OpenAI, Groq | deep-agent-ppb.streamlit.app

- Developed an advanced AI-powered conversational assistant utilizing LangChain and the DeepAgents framework, featuring a multi-agent Manager-Worker architecture to intelligently delegate complex tasks.
- Engineered persistent cross-thread memory using LangGraph checkpointers and a dynamic context retrieval system, enabling continuous memory retention and on-demand agent skill acquisition.
- Integrated live web search via Tavily API and enforced robust data exchange by utilizing Pydantic schemas to mandate strictly typed, structured JSON outputs within a responsive Streamlit UI.

Experience

AI/ML Intern – LLM Testing **Mar 2026 – Apr 2026**

Rigzz Technologies Kolkata, West Bengal

- Evaluated diverse LLM outputs by performing Human-in-the-Loop (HITL) comparative analysis, measuring the qualitative deviation between AI-generated responses and natural human baselines.
- Identified and documented edge-case behaviors, formatting errors, and unnatural phrasing to ensure high output accuracy prior to production releases.

Education

Vellore Institute of Technology **Aug 2023 – May 2027**

B.Tech in Computer Science and Engineering – CGPA: 7.74 Bhopal, India

Achievements

- **Patent:** Self-Recharging System for an Unmanned Aerial Platform
- **2nd Place Winner** – Innovation Hackathon for Mahakumbh MP 2028 Preparedness (Kumbh-Cortex Omni-Flow), Jun 2026

Extra Curricular

- **Operations Lead** – The Bengali Club, VIT Bhopal (Managed event workflows and logistics), Mar 2024 – Mar 2025
- **Core Member, Marketing** – Anterix Club, VIT Bhopal (Executed targeted campaigns to boost outreach), May 2025 – May 2026

Certifications

- Complete Data Science, ML, DL, & NLP Bootcamp – Udemy
- Complete Generative AI Course With Langchain and Huggingface – Udemy

Soft Skills

Teamwork & Collaboration · Problem-Solving · Adaptability · Time Management · Leadership & Mentoring